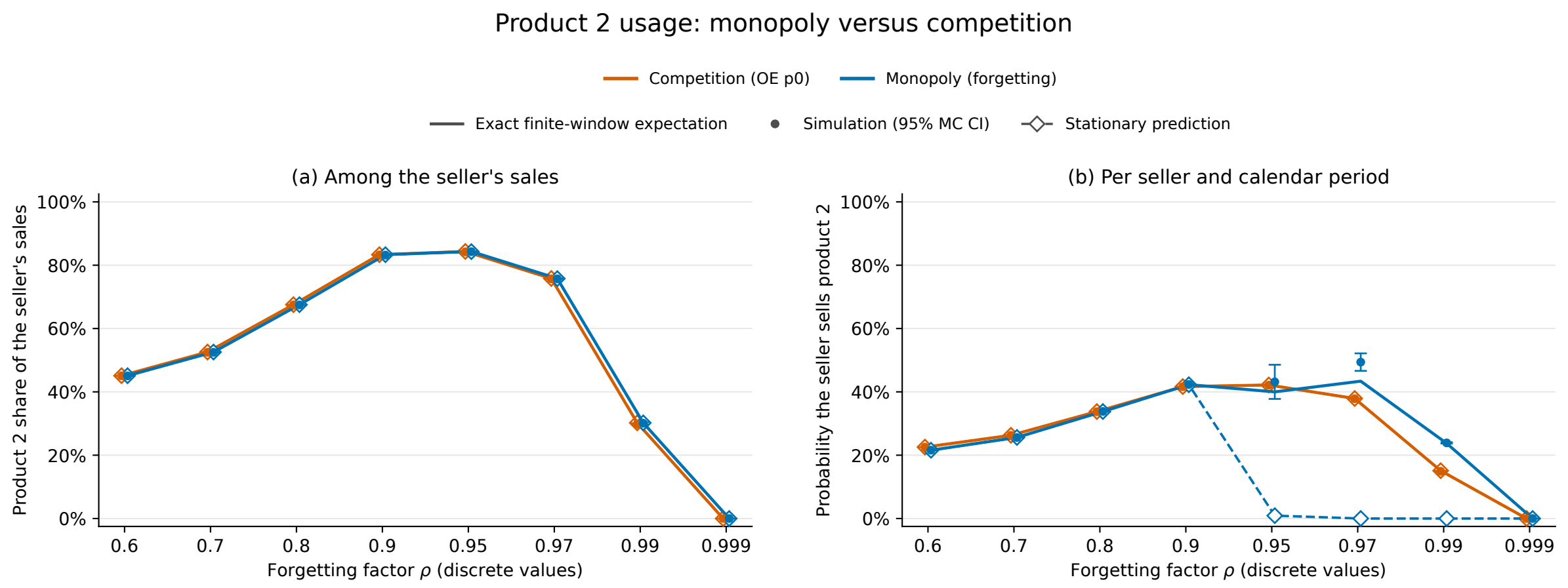
Product 2 usage: monopoly versus competition
Competition (OE p0)
Monopoly (forgetting)
Exact finite-window expectation
Simulation (95% MC CI)
Stationary prediction
(a) Among the seller's sales
(b) Per seller and calendar period
Product 2 share of the seller's sales
Probability the seller sells product 2
100%
80%
60%
40%
20%
0%
0.6
0.7
0.8
0.9
0.95
0.97
0.99
0.999
Forgetting factor ρ (discrete values)
Monopoly: one seller versus fixed $p_0^{OE}(\rho)$. Competition: two sellers following OE p0.
20 paths; 100,000 retained periods; 10,000 burn-in; initial reputation 0.5; common 51-point grid.
At high rho, finite-window and stationary predictions differ; 20-path MC intervals can also miss rare events.
Rho values are equally spaced; lines guide the eye. Stationary results refer to this finite grid.